

$H \rightarrow WW + \text{charm}$

Analysis status

Chirayu Gupta · VUB · August 2026

2022postEE · 26.7 fb⁻¹ · expected UL 1034

Sections

§	topic
1	Analysis and selection
2	MVA-defined regions
3	c-tagging — 2D working points and scale factors
4	Negative-weight reweighting
5	W+jets jet-binned samples
6	Systematics and impacts
7	Status and outlook

Dedicated decks: [ctag-sf-deck.pdf](#) (34 slides) · [negrw-deck.pdf](#) (57 slides)

1 · Analysis and selection

Why H+c

The Higgs coupling to **charm** is the least constrained of the accessible Yukawas. Associated production **H + c** probes it directly, without relying on the $H \rightarrow c\bar{c}$ decay.

production	$pp \rightarrow H + c (+X)$
decay	$H \rightarrow WW^* \rightarrow 2\ell 2\nu$
final state	opposite-sign $e\mu$ + MET + ≥ 1 c-tagged jet
dataset	2022postEE, 26.7 fb⁻¹ , ReReco 22Sep2023

The $e\mu$ requirement removes $Z \rightarrow ee/\mu\mu$ by construction, so the irreducible background is **real $t\bar{t}$** rather than Drell-Yan.

What $r = 1$ means. The signal template is normalised to the SM H+c prediction, **$\sigma \times B = 2.214 \times 10^{-3} \text{ pb}$** for $H \rightarrow WW \rightarrow 2\ell 2\nu$. A limit of **$r = 1034$** therefore excludes $\sim 1000\times$ the SM rate at this luminosity.

Signal and final state

H → WW → 2ℓ2ν, produced with a charm quark.

Final state: **opposite-sign eμ + MET + ≥1 c-tagged jet**

object	selection
muons	tight ID + tight PF iso, $p_T > 10$, $ \eta < 2.4$
electrons	wp80iso, $p_T > 10$, $ \eta < 2.5$, $\Delta R(e,\mu) > 0.4$
ll pair	OS, $p_T > 20 / 10$, pair $p_T > 30$, $m_{ll} > 12$
jets	$p_T > 30$, $ \eta < 2.4$, tight-lep veto ID, $\Delta R(j,\ell) > 0.4$
c-jets	$p_T > 20$, PNet medium CvL/CvB
MET	PuppiMET

eμ removes the $Z \rightarrow ee/\mu\mu$ peak **by construction** — the irreducible background is **real t $\bar{t}$** , not Drell-Yan. Triggers: MuonEG, SingleMu, SingleEle.

The selection is brutal on V+jets

process	SR yield	share of SR
$t\bar{t}$	17,744	82.3%
st	1,543	7.2%
vjets	1,523	7.1%
diboson	609	2.8%
higgsbkg	132	0.6%
H+c signal	0.26	0.001%
total	21,552	

The signal is **0.26 events** against ~21,600 background. Everything in this analysis follows from that ratio — which is why MC statistics, not data statistics, has been the limiting factor.

2 · MVA-defined regions

Six classes, one network

setting	value
model	SimpleMLP_MultiClass (b-hive)
classes	hplusc, higgsbkg, tt, st, diboson, vjets
features	26 , including the 11 c-tag one-hot categories
training	30 epochs, batch 1024, lr 1e-3, loss weighting on
split	80/20, deterministic by NanoAOD event id

The 11 one-hot categories mean the network sees the **full 2D CvL/CvB plane** — the same information the scale factors are binned in, not a binary pass/fail.

argmax defines the regions

The winning class picks the channel; the winning score is the discriminant.

region	events	dominant process
SR_hplusc	21,552	tt 82.3%
CR_higgsbkg	9,220	tt 86.4%
CR_tt	44,199	tt 94.1%
CR_st	20,135	tt 87.6%
CR_diboson	6,585	tt 72.4%
CR_vjets	9,214	vjets 43.2%

Regions are **orthogonal by construction** — no cut optimisation to tune or defend, and every event is used somewhere.

Four of five CRs are tt-dominated — by design

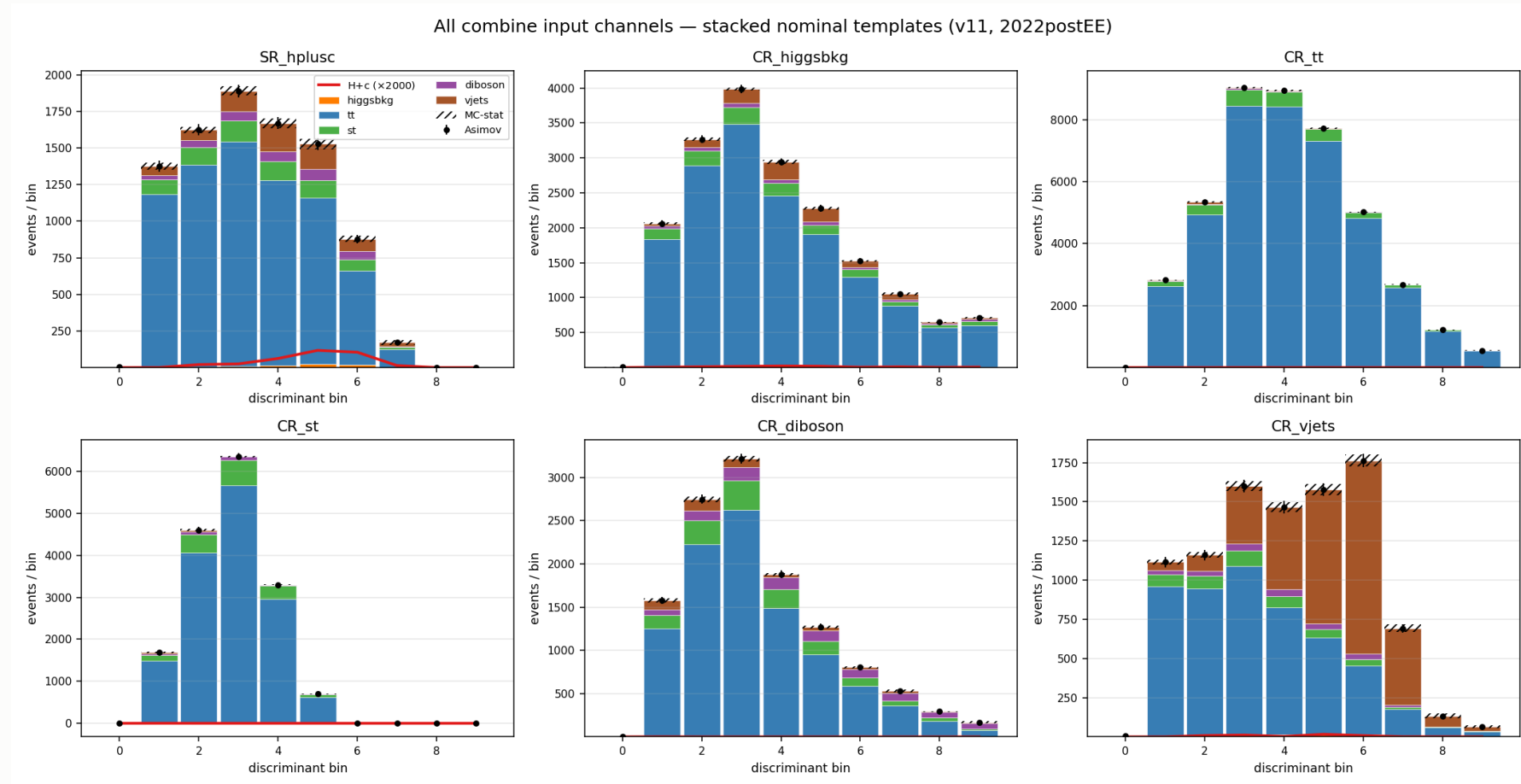
The SR is itself **82% tt**: in an ep final state $t\bar{t}$ is dominant everywhere, so a tt-rich CR is expected, not a failure.

CR_tt: 94.1% purity over 44k events. It pins the free-floating `rate_tt`, which covers 82% of the SR — the single most valuable constraint in the fit.

CR_vjets holds 59.6% of all V+jets in the fit. Without it there is no V+jets constraint from anywhere.

Collapsing the CRs to single bins was **measured**: +33 units for three CRs, ~+50 for all five. The shapes carry real constraint, so all six channels keep 10 bins.

Templates entering the fit



6 channels $\times$ 6 processes $\times$ 10 bins. **CR_vjets** (bottom right) is the only region where V+jets (orange) is visible — it carries 59.6% of all V+jets in the fit.

3 · c-tagging

From working points to a plane

PNet gives **two** discriminants per jet: **CvL** (charm vs light) and **CvB** (charm vs bottom). A single working-point cut throws most of that away.

The calibration instead partitions the **whole plane** into **11 categories** L0, C0–C4, B0–B4 — including the untagged L0 bin.

property	value
axes	CvL, CvB — verified sufficient , no third variable needed
categories	11: L0, C0 – C4, B0 – B4
index	computed from CvL/CvB already stored per jet
applied	natively in the processor (CTag2DCorrector)

Because the scheme spans the **full plane including L0**, the SF machinery works even for a selection with no working point at all.

Only 7 of 11 categories are populated

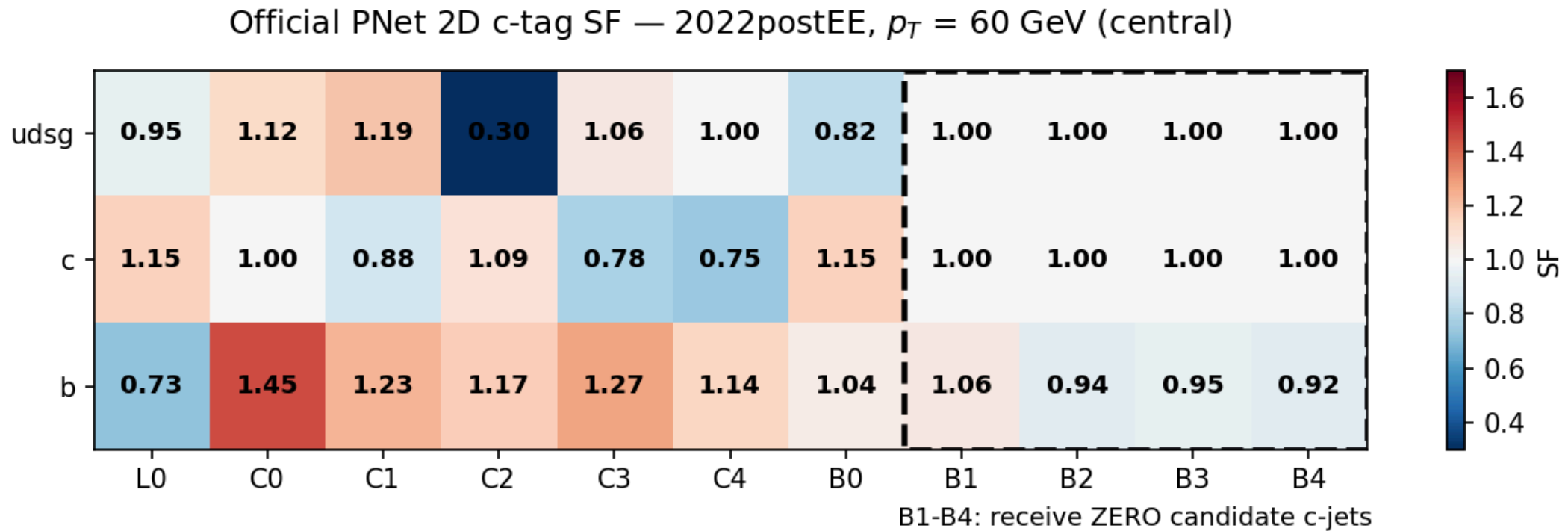
Occupancy of the five largest (C4 and B0 are populated but sparse):

category	jets	c (%)	b (%)	light (%)
L0	130,694	13.5	4.7	81.8
C0	188,702	28.2	6.2	65.7
C1	170,043	58.0	7.9	34.1
C2	17,823	58.6	33.4	8.0
C3	1,082	30.6	57.9	11.5

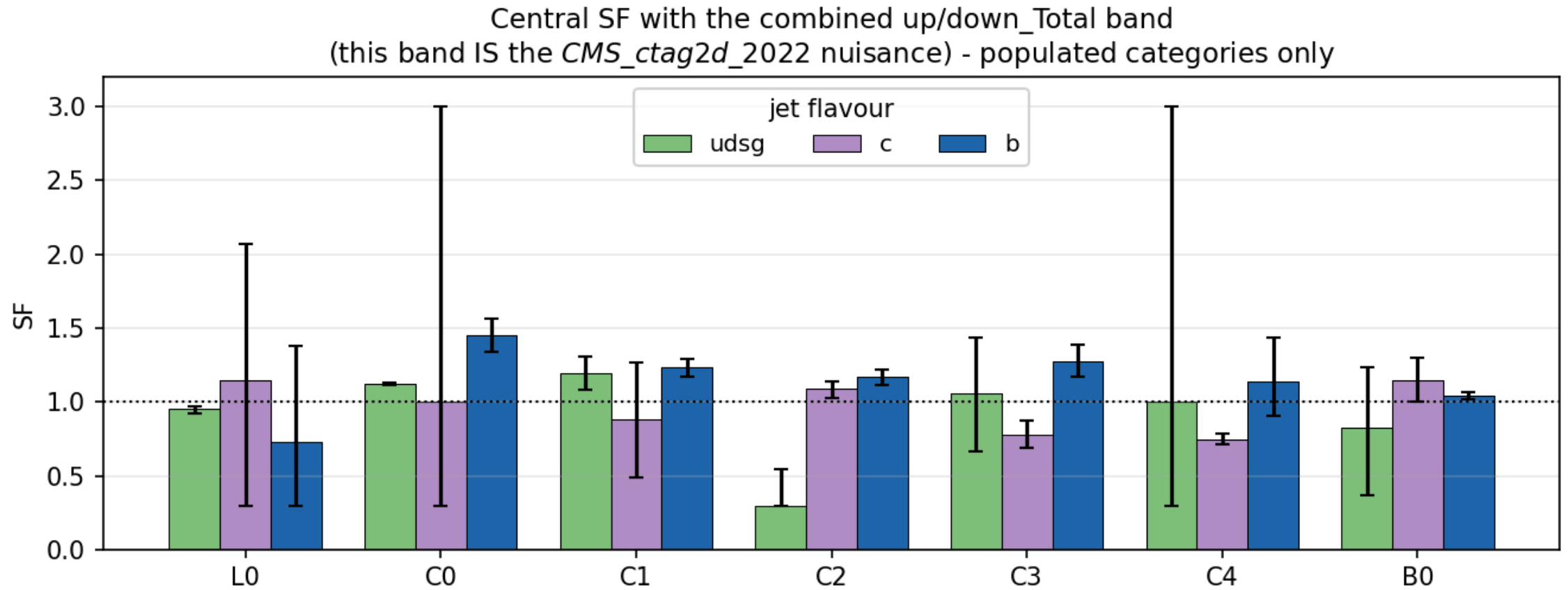
The scheme targets a broader phase space than ours — the b-rich B1 – B4 categories are empty after the $e\mu$ + c-jet selection.

Category index is computed from CvL/CvB already stored per jet — **no NanoAOD reprocessing was ever needed.**

The scale factor matrix



The uncertainty band is the nuisance



What the calibration costs

configuration	limit
no SF	1150
+ CMS_ctag2d_2022	1164

Applying the calibration **worsens** the limit by 14 units — as it must.
A scale factor adds a nuisance. The point is correctness.

Currently **one nuisance** covering the whole plane. Decorrelation is pending.

4 · Negative-weight reweighting

full detail: `negrw-deck.pdf` (57 slides)

The estimator is a cancellation

aMC@NLO events carry **signed** weights. The yield estimator

$$\hat{N}_B = \sum_{i \in B} w_i, \quad w_i = \pm |w_i|$$

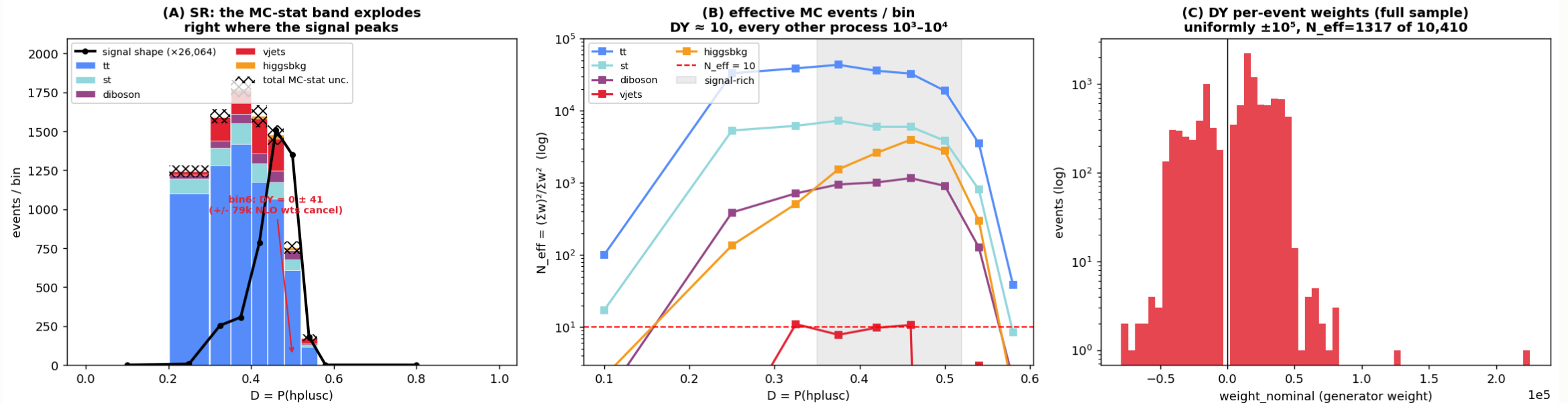
is a **difference of two large numbers**.

- the **expectation** is correct — the yield is unbiased
- the **variance** is not: it scales with $\sum w_i^2 = \sum |w_i|^2$, the *unsigned* statistics, while the mean is the much smaller signed sum
- effective statistics collapse: $N_{eff} = (\sum w)^2 / \sum w^2 \ll N$

16.4% of our V+jets events carried a negative weight. In a sparse bin the positives and negatives can cancel **completely** — leaving zero content with a large finite error.

V+jets was starved exactly under the signal

autoMCStats blow-up = DY (vjets) is starved in the H+c signal region



(A) the MC-stat band explodes where the signal peaks — one bin was $DY = 0 \pm 41$ from $\pm 79k$ weights cancelling · (B) N_{eff} per bin: V+jets ≈ 10 , every other process 10^3 – 10^4 · (C) DY per-event weights, uniformly $\pm 10^5$

The method — an algebraic identity

Write the signed generator density as a difference of two positive densities:

$$\text{PDF}(\vec{x}) = a \text{PDF}_+ - b \text{PDF}_-, \quad a - b = 1$$

With $P_+(\vec{x}) = a\text{PDF}_+ / (a\text{PDF}_+ + b\text{PDF}_-)$:

$$\text{PDF}(\vec{x}) = \underbrace{(2P_+(\vec{x}) - 1)}_{g(\vec{x})} \cdot (a\text{PDF}_+ + b\text{PDF}_-)$$

The right factor is the **unsigned** density — what $\sum |w|$ samples.

So $\sum |w| g$ and $\sum w$ estimate **the same distribution**. Yield-preserving by construction, not an approximation.

Exact for the *true* P_+ . We use $\hat{P}_+$ from a finite classifier — so closure becomes an **empirical** property that must be measured.

Training region — train loose, infer tight

$g(\vec{x})$ is a **generator property**; it does not know about the analysis selection.
So we train on a far larger sample than the SR.

	region
train	all base cuts + veto of the $e\mu$ SR topology
infer	the tight $e\mu$ signal region

Disjoint by construction. An event that is not "exactly one $e\mu$ pair" can never be an SR event — no event-id bookkeeping needed.

Rejected alternatives: inverting MET (kinematically softer $\rightarrow$ extrapolation) and inverting the c-jet requirement (flips into the *opposite* gen corner).

Inputs — 20 generator-level features

group	variables
LHE event	<code>lhe_njets</code> , <code>lhe_nb</code> , <code>lhe_nc</code> , <code>lhe_nuds</code> , <code>lhe_nglu</code> , <code>lhe_npnlo</code>
LHE kinematics	<code>lhe_ht</code> , <code>lhe_htincoming</code> , <code>lhe_vpt</code> , <code>lhe_alphas</code>
gen partons	multiplicity, <code>n_pt20/100/200</code> , incoming PDG IDs
leading partons	<code>genparton1/2_pt</code> , <code>genparton1/2_eta</code>

All generator-level. The paper explicitly **rejects reco-level variables** — closure is only guaranteed on quantities the generator itself sampled.

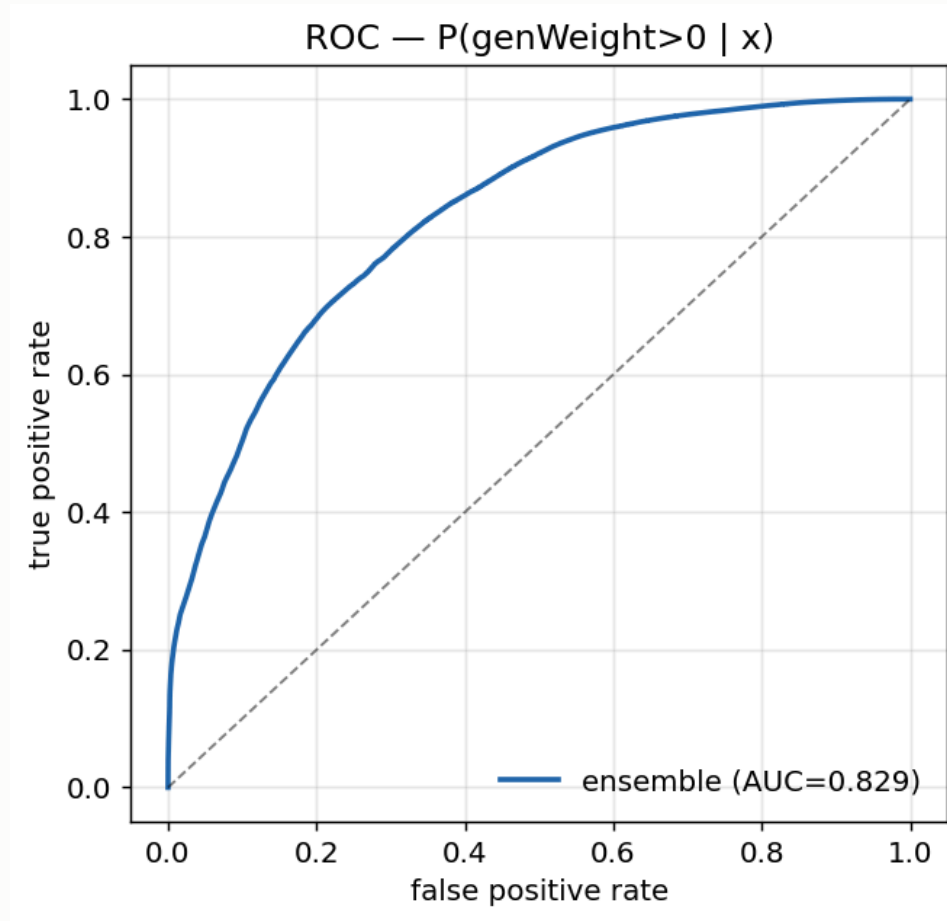
The two weight classes separate visibly in nearly every one of these features.

The classifier

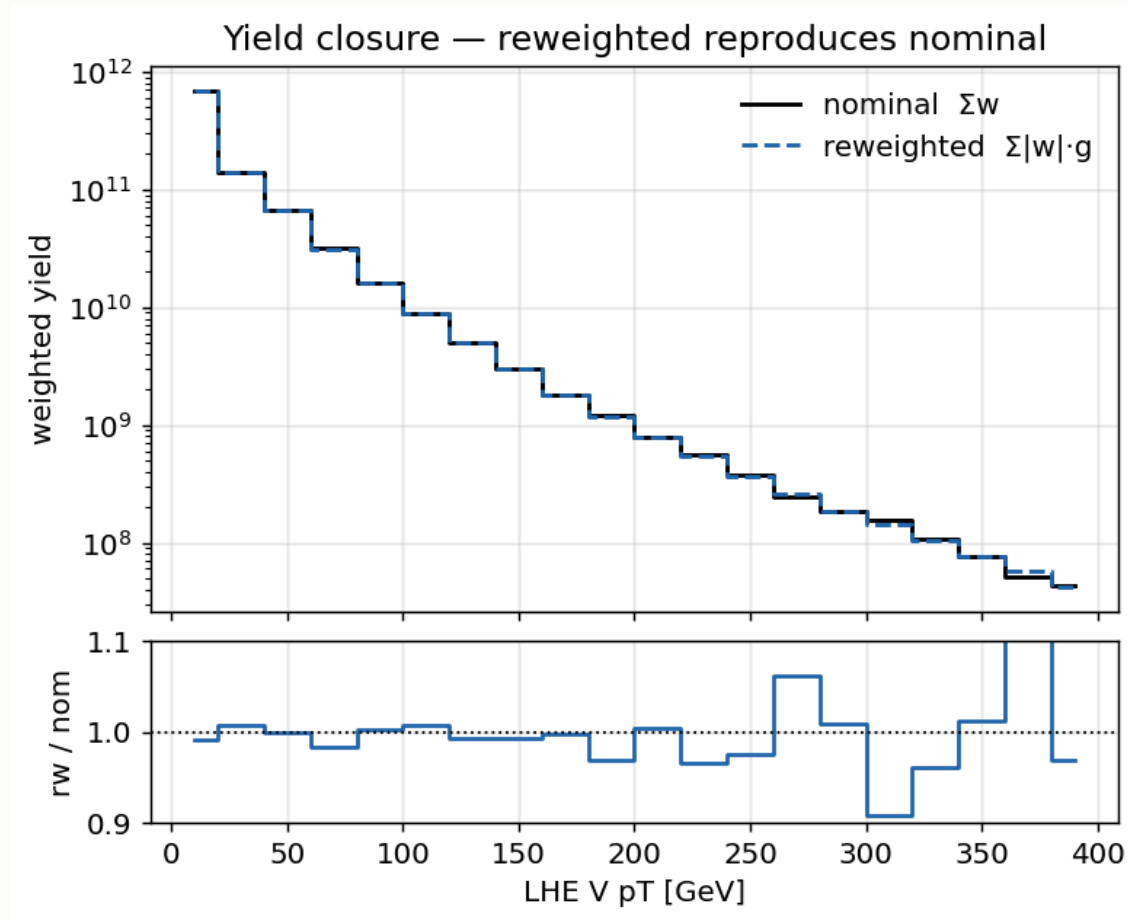
setting	value
model	HistGradientBoostingClassifier (scikit-learn)
inputs	20 generator-level (<code>lhe_*</code> , <code>genparton_*</code>)
training events	9.8M
ensemble AUC	0.829

AUC 0.83 on an **intrinsically stochastic** target — the weight sign is not a deterministic function of the kinematics, so a perfect classifier cannot exist even in principle. 0.83 means the generator-level features genuinely carry the information the NLO subtraction encodes.

Classifier ROC

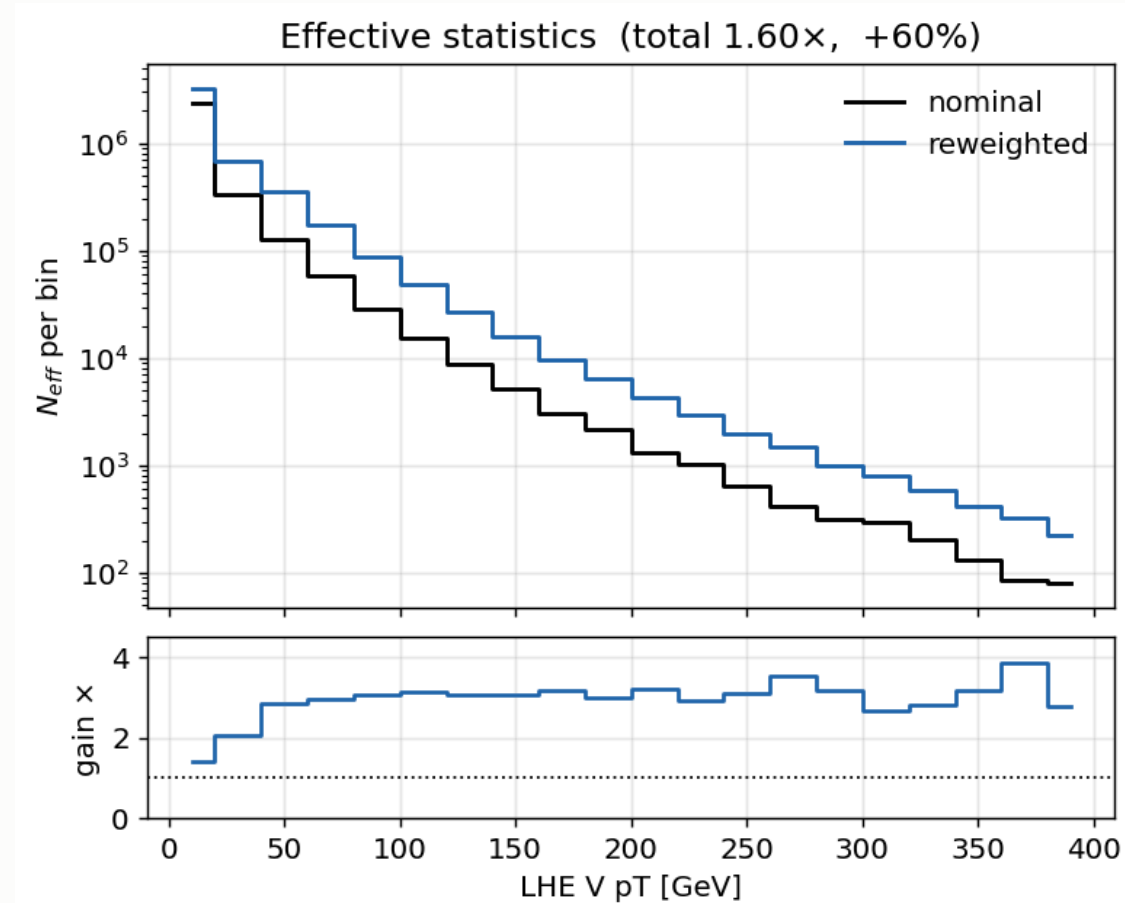


Closure



Training-region closure **0.994** on 9.8M events — the reweighting removes variance and nothing else.

The gain



Integrated gain **1.6×**; per-bin ratio $\approx$ **3×** in the signal-rich bins.

The honest caveat

The SR closure carries an offset. The signal region is a small, kinematically-biased corner of the space the classifier was trained on, so a finite $\hat{P}_+$ does not close perfectly there.

Why this is acceptable, stated plainly:

- reweighting fixes **variance, not yield** — exact closure is what the identity guarantees for a *perfect* classifier, not a finite one
- it **keeps the entire benefit**: N_{eff} comes from reduced per-bin variance, which survives the offset
- the residual is covered by `CMS_negrw_vjets`, which profiles to **0.0%**

This caveat is our extension, not the paper's — their closure was clean because their evaluation region was not this starved.

5 · W+jets jet-binned samples

Inclusive → jet-binned

AN-23-102 §2.3 rejects the inclusive NLO sample outright:

"not used... 5 times smaller than LO and with large fraction of negative weights."

sample	xsec (pb)	events	files	neg-w
WtoLNu-2Jets_0J	55,760	678M	3,432	10.2%
WtoLNu-2Jets_1J	9,529	523M	2,669	25.7%
WtoLNu-2Jets_2J	3,532	345M	2,135	34.7%
sum	68,821	1,546M	8,236	
<i>old inclusive</i>	<i>67,710</i>	<i>282M</i>	<i>381</i>	<i>16.1%</i>

Cross sections from **XSDB**; sum within **+1.6%** of the inclusive — normal NLO merging spread, not a double-count. Measured negative-weight fractions match XSDB to ~1%.

XSDB labels these accuracy: "L0" — **wrong**, auto-populated. `amcatnloFXFX` is NLO, proven by the 10–35% negative weights, impossible at LO.

Why the gain is not simply 5.5×

Raw events rise **5.5×**, but the useful quantity is effective statistics:

sample	n_{eff}/N	equivalent lumi
0J	0.635	7.7 /fb
1J	0.237	13.0 /fb
2J	0.094	9.1 /fb
combined		29.8 /fb
<i>inclusive</i>	<i>0.460</i>	<i>1.9 /fb</i>

The inclusive sample spends most of its cross section on **0-jet** events, which almost never survive the ≥ 1 c-jet requirement.

Of the 4,851 surviving SR events, **2J alone contributes 69%** — despite being the smallest sample. The jet-binned samples are enriched in exactly what the selection needs.

W+jets result: 1160 → 1034

	before	after
expected UL	1160	1034
V+jets n_{eff} (SR)	280	1170
V+jets stat. error	5.98%	2.92%
V+jets rate	1508	1523

The rate barely moved while the statistical error halved — the signature of a statistics gain, not a normalisation change.

Was the gain really W+jets?

tt (+4.8%) and st (+3.8%) also moved between the two cards — both were re-merged from scratch. Checked per-process:

process	n_{eff} ratio
vjets (SR)	4.18×
vjets (CR_st)	9.89×
vjets (CR_tt)	5.86×
tt, st, diboson, higgsbkg	0.87 – 1.13×

The n_{eff} gain is confined to V+jets. Other processes move by $\leq 13\%$ in *both* directions — re-merge jitter, not a systematic shift. Both cards were also re-verified independently: **1160** and **1034**.

6 · Systematics

The full inventory

22 shape · 9 lnN · 1 rateParam · autoMCStats

shape (weight)	shape (object shift)
pileup	CMS_scale_j_2022 (JES)
ps_isr, ps_fsr	CMS_res_j_2022 (JER)
scalevar_muR, _muF, _muR_muF	CMS_scale_e_2022, CMS_res_e_2022
lhe_pdf, lhe_alphaS	CMS_scale_m_2022, CMS_res_m_2022
muon_id, muon_iso	
electron_id, electron_reco ×3	lnN
CMS_ctag2d_2022	lumi_13p6TeV, xsec_st/diboson/vjets/higgsbkg
CMS_negrw_vjets	BR_HtoWW, BR_Htautau, xsec_hplusc_4FS_5FS
flavor_composition_ggH (placeholder — see §Higgs heavy-flavour)	

Plus **rate_tt** (free rateParam, tt from data) and **autoMCStats** (threshold 10)

Object shifts need full reprocessing

JES/JER and lepton scale/resolution **change the MVA score**, so they cannot be applied as an event weight.

Each is a **separate parquet tree**, re-run through the full selection *and* re-scored by the network — 12 shifted directories.

This is the trap that once cost 500 units. A partial inference run scored only 19 of 57 directories and left object-shift templates frozen at nominal; the limit read 1676 instead of 1185. Inference coverage is now verified after every rebuild.

What we fixed — 1 · top- p_T reweighting

Our first implementation was **wrong** and was corrected against `hh2bbww` (the Hamburg framework behind AN-24-091).

	what we had	what we now apply
form	data-based exponential	theory-based (NNLO/NLO)
Run 3	none	$\times (0.991 + 7.5e-5 \cdot p_T)$
nominal	left at 1.0	reweighted
p_T cap	invented 500 GeV	none (form is well-behaved)

```
sf_run2 = 0.103*exp(-0.0118*pT) - 0.000134*pT + 0.973
sf       = (0.991 + 0.000075*pT) * sf_run2
weight   = sqrt(prod(sf))           # over the two gen tops
down     = 1.0                     # "no correction" is the variation
```

Our original coefficients were real — but belonged to the *other*, data-driven variant.

What we fixed — 2 · Higgs heavy-flavour

The original **flat lnN on merged higgsbkg was mis-scoped**: ggH is only **13.1%** of that group (VBF 29.0%, ggZH 23.3%, ZH 21.0%, WH 9.1%).

A flat lnN either over-penalises the other 87% (1.40) or must be watered down to an average (1.066) — and **cannot produce a shape effect at all**.

Replaced with a per-event weight, ported from HiggsDNA `Higgs_plus_HF_syst`:

```
num_HF_jets = ak.sum(genJets.hadronFlavour == 4, axis=-1) # pT>25, |eta|<2.5
up = where(num_HF_jets > 0, 1.5, 1.0) # AN-23-102: 50% on ggH
down = where(num_HF_jets > 0, 0.5, 1.0)
```

Keys on **gen-jet flavour**, so process grouping stops mattering — and it produces the shape a flat lnN structurally cannot.

What we fixed — 3 · MET unclustered energy

Implemented, not yet in the card — it needs the shifted trees from a reprocessing pass, so it does not appear in the 22 shape nuisances above.

The `CorrectedMETFactory` path **never runs for Run 3 PuppMET**, so the unclustered-energy shift was silently absent.

Now taken directly from the NanoAOD branches:

```
events.PuppMET.ptUnclusteredUp / ptUnclusteredDown  
events.PuppMET.phiUnclusteredUp / phiUnclusteredDown
```

Independently confirmed against HiggsDNA `MET_syst_Unclustered` — same branches, same up/down ordering.

What we added

systematic	what was done
CMS_negrw_vjets	new — ensemble spread of the negative-weight reweighting
CMS_ctag2d_2022	new — 2D CvL/CvB SF applied natively in the processor
electron_reco ×3	split into p_T bins (<20, 20–75, >75 GeV)
lhe_pdf, lhe_alphaS	added — NNPDF replicas, MC2Hessian
top_pt	rewritten to the hh2bbww theory form — <i>pending activation</i>
higgs_plus_c	new per-event HF weight — <i>pending activation</i>
MET unclustered	new object shift — <i>pending activation</i>

The last three are implemented and verified but **not in the current card** — each needs its weight column or shifted tree from a reprocessing pass.

What we checked and deliberately excluded

item	decision
muon reco SF	not applicable — HiggsDNA exposes no reco key; hh2bbww registers <code>mu_id_sf</code> / <code>mu_iso_sf</code> and no <code>mu_reco_sf</code> . Two frameworks, same conclusion.
PU jet ID	absent from both frameworks for Run 3
UE / tune	requires dedicated tune samples — cannot be a weight
<code>hdamp</code> , <code>mtop</code>	sample-based tt modelling; absent from AN-23-102 Table 16 too

`hdamp` and `mtop` are standard Run 3 tt modelling terms. Both need alternative samples. Recorded as a **documented decision**, not an oversight.

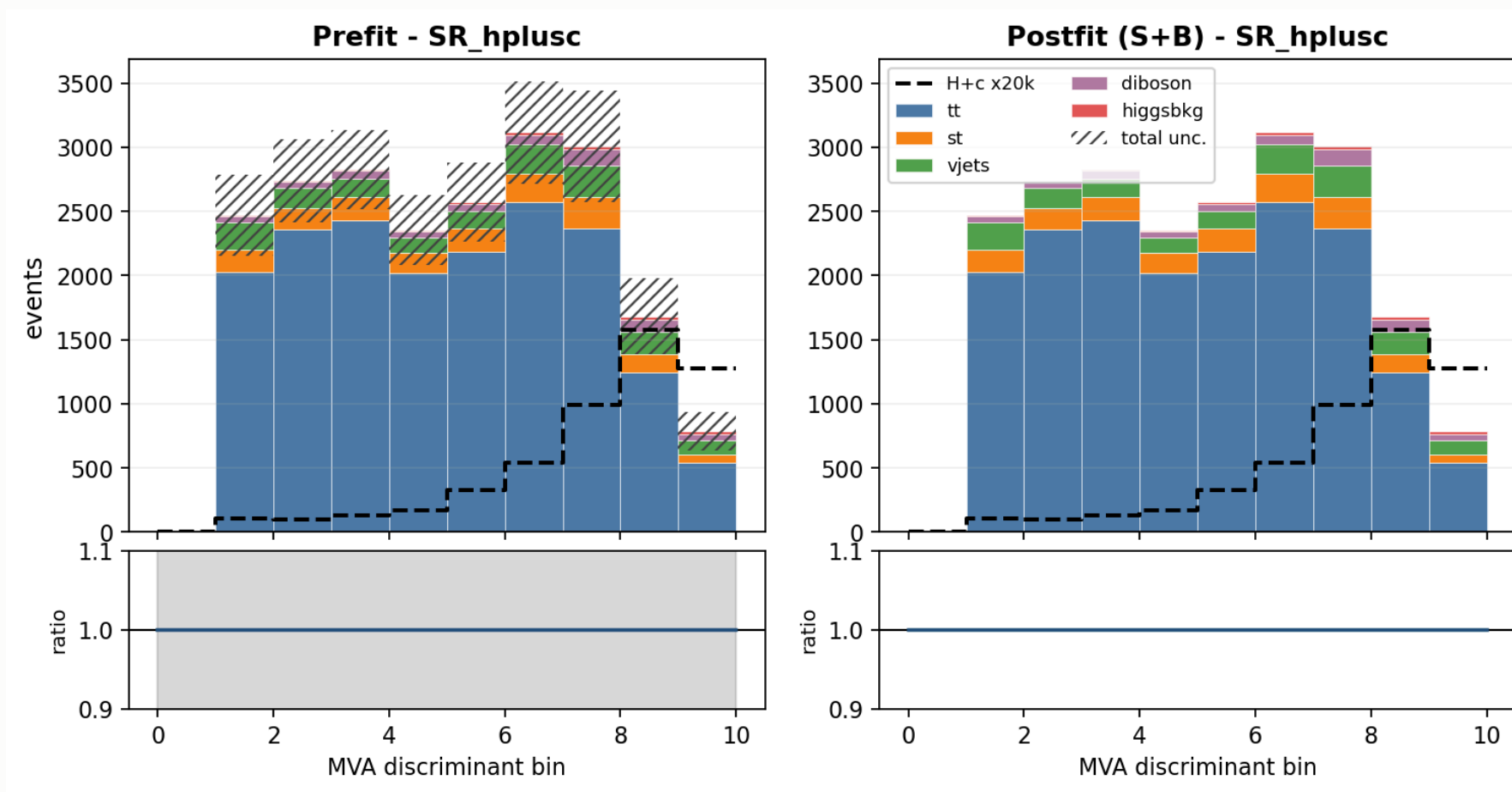
What moves the limit

Each nuisance frozen in turn; **Δ is the improvement in the expected limit.**

frozen	limit	Δ	% of 1034
all constrained (stat-only)	641	393	38.0%
theory shapes (scale + PS + α_S)	883	151	14.6%
xsec_hplusc_4FS_5FS	921	113	10.9%
scalevar_muF	944	90	8.7%
autoMCStats (all)	956	78	7.5%
CMS_ctag2d_2022	964	70	6.8%
rate_tt	1011	23	2.2%
JES	1023	11	1.1%

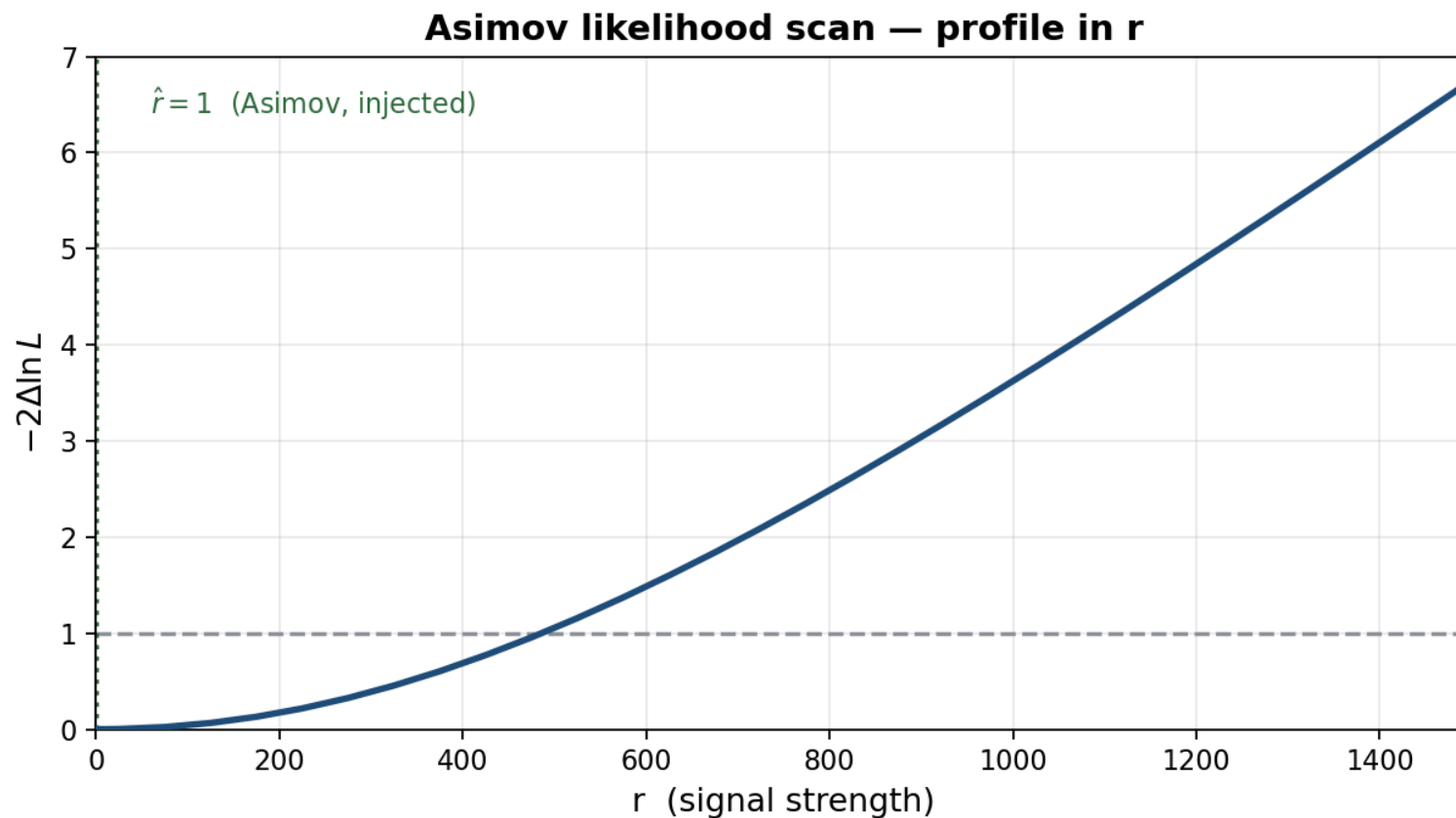
autoMCStats is no longer the leading term. It was dominant before the W+jets change; at 7.5% the analysis is no longer MC-stat-limited in the way it was. Signal theory now leads.

Prefit vs postfit — signal region



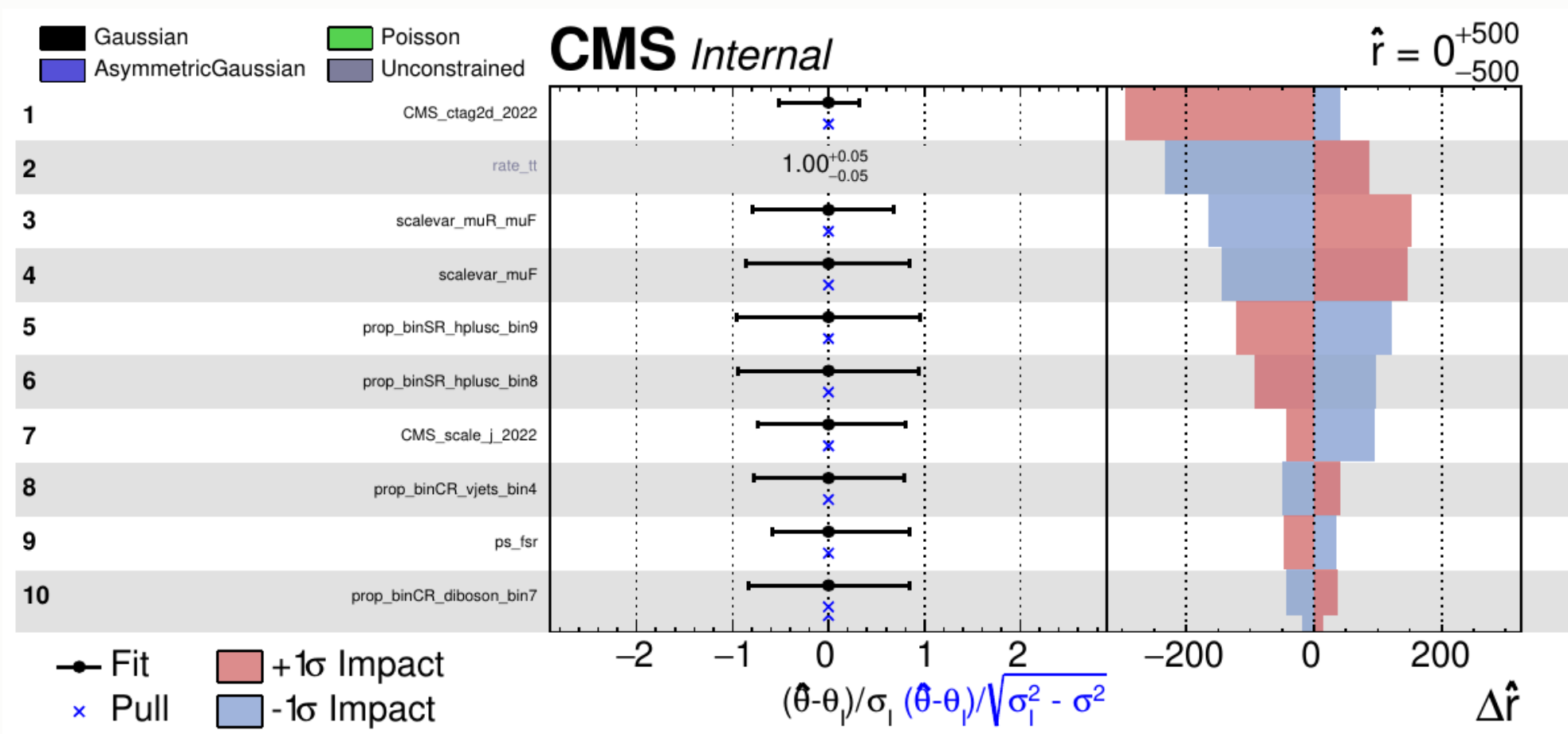
The prefit uncertainty band (hatched) collapses after the fit — that is the constraint the control regions provide. The ratio panel makes it explicit: prefit band $\pm 10\%$, postfit a line. Asimov $r = 1$, so the ratio is 1 by construction;

Likelihood scan



Minimum at the injected $\hat{r} = 1$. The dashed line is the 68% CL crossing;
the quoted **1034 is a CLs upper limit**, which is a different construction and is
not read off this curve.

Nuisance impacts — top 10



The header $\hat{r} = 0^{+500}_{-500}$ is a **plotImpacts rounding artefact** — the actual fit returns $\hat{r} = 0.99996, -513/+483$, i.e. exactly the injected Asimov value.

Reading the impacts — the asymmetries

Left panel — the pull $(\hat{\theta} - \theta_0)/\sigma_\theta$. All pulls sit at zero:

this is an **Asimov** fit, so by construction nothing is pulled. A non-zero pull here would mean the fit is being dragged, and would need explaining.

Right panel — $\Delta\hat{r}$ when each nuisance is moved by $\pm 1\sigma$.

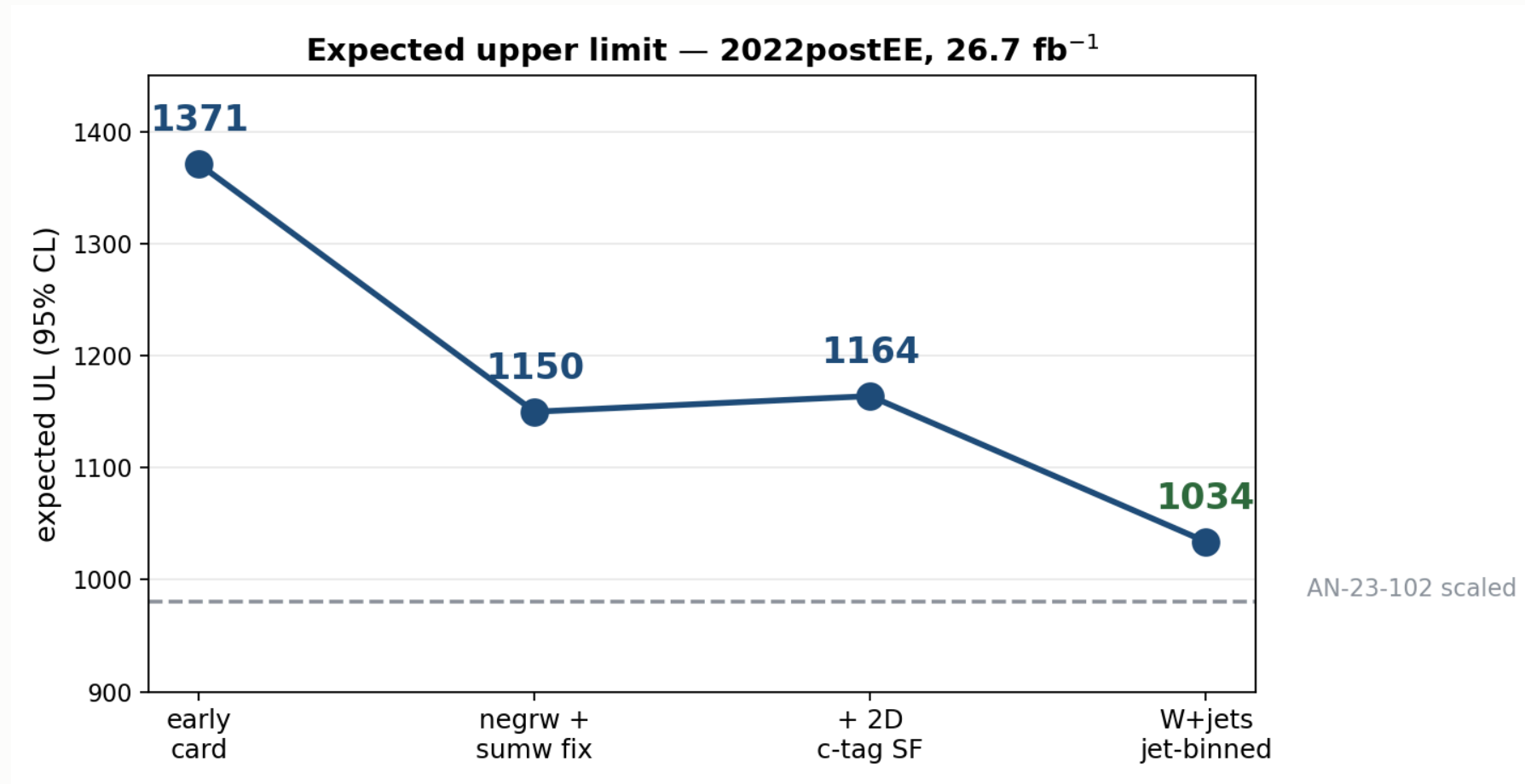
Red (+1 σ) and blue (−1 σ) are not mirror images. Three sources of asymmetry:

source	why
asymmetric lnN	<code>xsec_st</code> is <code>0.9873/1.0167</code> — the up and down variations differ by construction
template shape	up/down shifts of JES or ctag move events between bins non-linearly
re-profiling	pushing one nuisance lets the others re-fit, and they respond differently in each direction

The **fit range matters**. An earlier run with `--rMin 0` clipped every -1σ impact at the boundary and produced a one-sided plot. A symmetric range around $\hat{r}$ is required for the impacts to be meaningful.

7 · Status and outlook

The road so far



Note the suppressed zero on the y-axis — the c-tag step is a genuine but small **+14** increase, not the spike it appears to be.

Where we stand

	limit
AN-23-102 scaled to 26.7 fb ⁻¹	980
this analysis	1034
our statistical only	641

Within **~5%** of the published analysis scaled to our luminosity, on **5× less data**.

The scaling is indicative, not rigorous. $431 \times \sqrt{138/26.7} = 980$ assumes a purely statistics-limited extrapolation. AN-23-102's own Table 17 puts its statistical term at 73.8%, so it is *not* fully statistics-limited and the true scaled value would be somewhat worse than 980 — i.e. our gap is likely **smaller** than 5%, not larger.

The systematic breakdown of the 1034 is **under investigation** — the full term-by-term comparison against AN-23-102 Table 17 is not yet settled.

What is still missing

No data/MC agreement plots yet — the analysis is blind and the comparison plots are still being produced.

item	status
4FS/5FS signal theory	placeholder 1.30 — needs re-derivation at 13.6 TeV
Trigger scale factors	implemented, not yet enabled
Higgs heavy-flavour (<code>higgs_plus_c</code>)	implemented, pending activation
top-p_T reweighting	implemented, pending activation
Dataset substitutions	W+jets p_T -binned (full AN stitch)
	DY $\rightarrow$ Z $\rightarrow$ $\tau\tau$ filtered
	tt / single-top <code>-ext</code> samples
	<code>WZto3LNu</code>

Priorities

#	item	why
1	Re-derive <code>xsec_hplusc_4FS_5FS</code>	30% flat lnN on a statistics-starved signal
2	Enable trigger SFs	implemented, one flag
3	Activate <code>higgs_plus_c</code> + <code>top_pt</code>	proper HF and tt treatment
4	Add the missing datasets	more MC statistics
5	Decorrelate <code>CMS_ctag2d_2022</code>	currently one nuisance over the whole plane

Summary

Expected UL 1371 → 1034 · statistical-only 641

V+jets n_{eff} **280 → 1170**

Within ~5% of the Run 2 analysis scaled to our luminosity

Remaining: 4FS/5FS re-derivation, trigger SFs, dataset substitutions